

Total No. of Questions : 8]

SEAT No. :

PC2803

[6352]-27

[Total No. of Pages : 3

S.E. (Electronic/ E&Tc) (Electronics & Computer/VLSI Design & Technology/Electronics & Communication/Advanced Communication Technology)
DIGITAL CIRCUITS (OOP)
(2019 Pattern) (Semester-III) (204182)

Time : 2 ½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.No.1 or Q.No.2, Q.No.3 or Q.No.4, Q.No.5 or Q.No.6, Q.No.7 or Q.No.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicates full marks.

Q1) a) Explain the working of a half-adder? Draw its logic diagram. [7]

b) Implement the full subtractor using a 1:8 demultiplexer. [5]

c) Implement the following function using multiplexer [5]

$$f(A,B,C) = \Sigma m(0,2,4,6)$$

OR

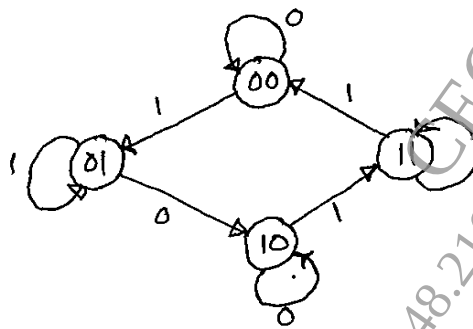
Q2) a) Draw the logic diagram of full-adder and its truth table. [7]

b) Implement a full-adder using Demultiplexer. [5]

c) Implement the given logic function using a 4:1 multiplexer [5]

$$f(A,B,C) = \Sigma m(0,2,4,6)$$

Q3) a) For the state diagram shown in figure, obtain the state table and design the circuit using minimum number of J-K flip-flops. [8]



P.T.O.

- b) Explain the function of a shift register. Give its application. [5]
- c) Explain with truth table the working of clocked RS flip-flop. [5]

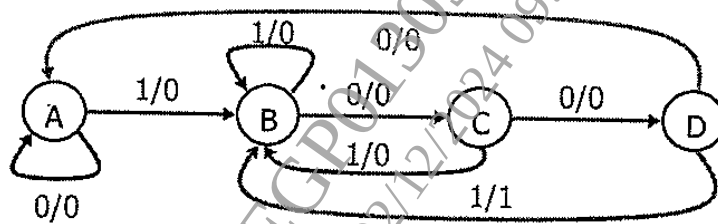
OR

- Q4)** a) Design a sequence generator using T FFs. [8]



- b) Explain the types of shift register. [5]
- c) Explain with diagram the working of D type Flip-flop. Give its truth table. [5]

- Q5)** a) Design the clocked sequential circuit for the state diagram using JR flip flop. [9]



- b) Draw ASM chart for a 2 bit up- down counter having mode control input M. [8]

M= 1 Up counter.

M= 0 Down counter.

OR

- Q6)** a) Design a sequential circuit using Mealy machine for detecting the sequence....1001..... Use JK Flip-flop. [9]

- b) Explain in short: [8]

- State Diagram
- ASM chart

Q7) a) Explain the classification of memories based on their principle of operation. [8]

b) Write a short note on concept of PLA and PAL. [10]

OR

Q8) a) Explain with circuit diagram the dynamic MOS memory. [8]

b) A combinational circuit defined by the function [10]

$$F_1(A,B,C) = \sum (3,5,6,7) \text{ and}$$

$$F_2(A, B, C) = \sum (0,2,4,7)$$

Implement the circuit with PLA having 3 inputs, 3 product terms and 2 outputs.

